

Week 5: aquatic invertebrates

Individual assignment 5

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2026-05-10

1. Set up

Packages

```
#Loading general use and cleaning packages
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Loading package for wrapping long axis labels
library(scales)

#Loading package for reading Excel files
library(readxl)

#Loading package for visualizing missing data
library(naniar)

#Loading package for arranging plots
library(patchwork)
```

Data

```

#Reading in taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

#Reading in aquatic invertebrate data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

#Reading in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                  sheet = "sites")

#Reading in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪ Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))

```

Cleaning

```

#Creating a new object from site information
ncos_sites <- sites |>
  #Filtering to only include NCOS sites sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

#Creating a clean taxon list for joining with aquatic invertebrate data
taxon_list_clean <- taxon_list |>
  #Converting taxon names to snake case so they match the aquatic
  ↪ invertebrate columns
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

```

#Creating a clean water quality object
water_quality_clean <- water_quality |>
  #Cleaning column names into snake case
  clean_names() |>
  #Keeping only water quality surveys from quarterly NCOS sites
  semi_join(ncos_sites, by = "site") |>
  #Creating a date-site column for joining with aquatic invertebrate data
  unite("date_site", date, site, remove = TRUE) |>
  #Grouping by date-site survey

```

```

group_by(date_site) |>
#Calculating median pH, salinity, and dissolved oxygen for each survey
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
#Ungrouping the dataframe
ungroup() |>
#Filtering out the salinity outlier identified in class
filter(date_site != "2022-08-12_NVBR")

```

```

#Creating a clean aquatic invertebrate object
aq_ins_clean <- aq_ins |>
#Cleaning column names into snake case
clean_names() |>
#Keeping only sites that occur in the NCOS sites object
semi_join(ncos_sites, by = c("site")) |>
#Renaming the vial date column to date
rename(date = date_on_vial) |>
#Selecting survey information and focal aquatic invertebrate taxa
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
#Keeping only filtered beaker samples
filter(sample_type %in% c("FB 250", "FB250")) |>
#Converting taxon columns into long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
#Grouping by date, site, and taxon
group_by(date, site, taxon) |>
#Calculating average abundance per liter from the 7.5 liter sample volume
summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
#Ungrouping the dataframe
ungroup() |>
#Creating a date-site column for joining with water quality data

```

```

unite("date_site", date, site, remove = FALSE) |>
#Joining water quality information to each survey
left_join(water_quality_clean, by = c("date_site")) |>
#Joining taxonomic information to each taxon
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
#Adding full site names for clearer figure labels
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
#Ordering sites by median salinity so site patterns are easier to compare
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪  = TRUE),
  site_full = fct_rev(site_full))

```

Problems

1. Visualizing differences across sites in major

taxon abundance

a. Plan your figure

- X-Axis: `site_full`
- Y-Axis: $\log + 1$ transformed average Ostracod abundance per liter
- Geometry to represent average abundance/L: `geom_jitter()`
- Geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
#Creating a new object from clean aquatic invertebrate data
ostracods <- aq_ins_clean |>
  #Filtering to only include ostracods
  filter(taxon == "ostracod") |>
  #Log + 1 transforming average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

#Displaying 10 random rows from the ostracod data frame
ostracods |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 24
```

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2023-05-04_NMC	2023-05-04 00:00:00	NMC	ostr~	1.87	8.32	NA	10.3
2	2023-09-12_NVBR	2023-09-12 00:00:00	NVBR	ostr~	0	9.14	NA	20.3
3	2023-02-28_NPB1	2023-02-28 00:00:00	NPB1	ostr~	0	NA	NA	NA
4	2022-05-11_NDC	2022-05-11 00:00:00	NDC	ostr~	0	9.11	0.398	9.4
5	2021-11-23_NPB2	2021-11-23 00:00:00	NPB2	ostr~	0.267	7.59	1.99	0.65
6	2022-08-15_NMC	2022-08-15 00:00:00	NMC	ostr~	0.133	7.61	NA	NA
7	2023-05-05_NPB	2023-05-05 00:00:00	NPB	ostr~	24.7	NA	NA	12.1
8	2021-11-12_NPB	2021-11-12 00:00:00	NPB	ostr~	79.2	NA	NA	NA
9	2022-02-23_NPB1	2022-02-23 00:00:00	NPB1	ostr~	0	NA	NA	NA
10	2023-05-06_NDC	2023-05-06 00:00:00	NDC	ostr~	56	10.1	NA	13.1

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

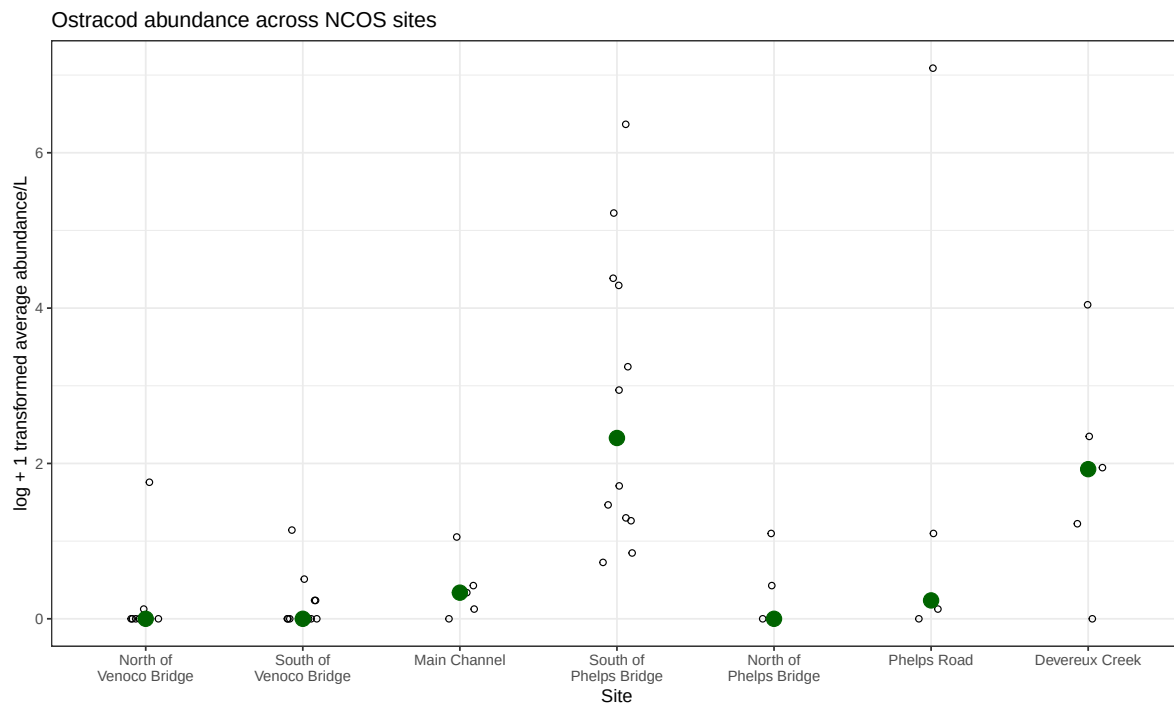
```
#Creating an ostracod abundance figure across sites
ostracods_plot <- ggplot(data = ostracods,
  #Setting x-axis to site and y-axis to
  ↪ log-transformed abundance
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  #Adding jittered points to show individual observations
  geom_jitter(height = 0,
    shape = 21,
```

```

        width = 0.1) +
#Adding larger points to show median ostracod abundance at each site
stat_summary(geom = "point",
             fun = "median",
             size = 4,
             color = "darkgreen") +
#Wrapping long site names on the x-axis
scale_x_discrete(labels = label_wrap(15)) +
#Relabeling axes and adding a title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracod abundance across NCOS sites") +
#Using a different theme from default
theme_bw()

#Displaying ostracod abundance figure
ostracods_plot

```



d. Create a multipanel figure

```
#Creating salinity figure from clean aquatic invertebrate data
salinity <- ggplot(data = aq_ins_clean,
                  #Setting x-axis to site and y-axis to median salinity
                  mapping = aes(x = site_full,
                                y = med_sal)) +

#Adding jittered points to show salinity values from each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +

#Wrapping long site labels on the x-axis
scale_x_discrete(labels = label_wrap(width = 15)) +

#Adding red points to show median salinity at each site
stat_summary(fun = median,
            color = "red",
            size = 1) +

#Relabeling axes and adding a title
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +

#Using a different theme from default
theme_bw()

#Creating a clean copepod object from aquatic invertebrate data
copepods <- aq_ins_clean |>
#Filtering to only include copepods
filter(taxon == "copepod") |>
#Log + 1 transforming average abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

#Creating copepod abundance figure across sites
copepods_plot <- ggplot(data = copepods,
                       #Setting x-axis to site and y-axis to log-transformed
                       ↪ abundance
                       mapping = aes(x = site_full,
                                       y = log_ave_lit)) +

#Adding jittered points to show individual observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +

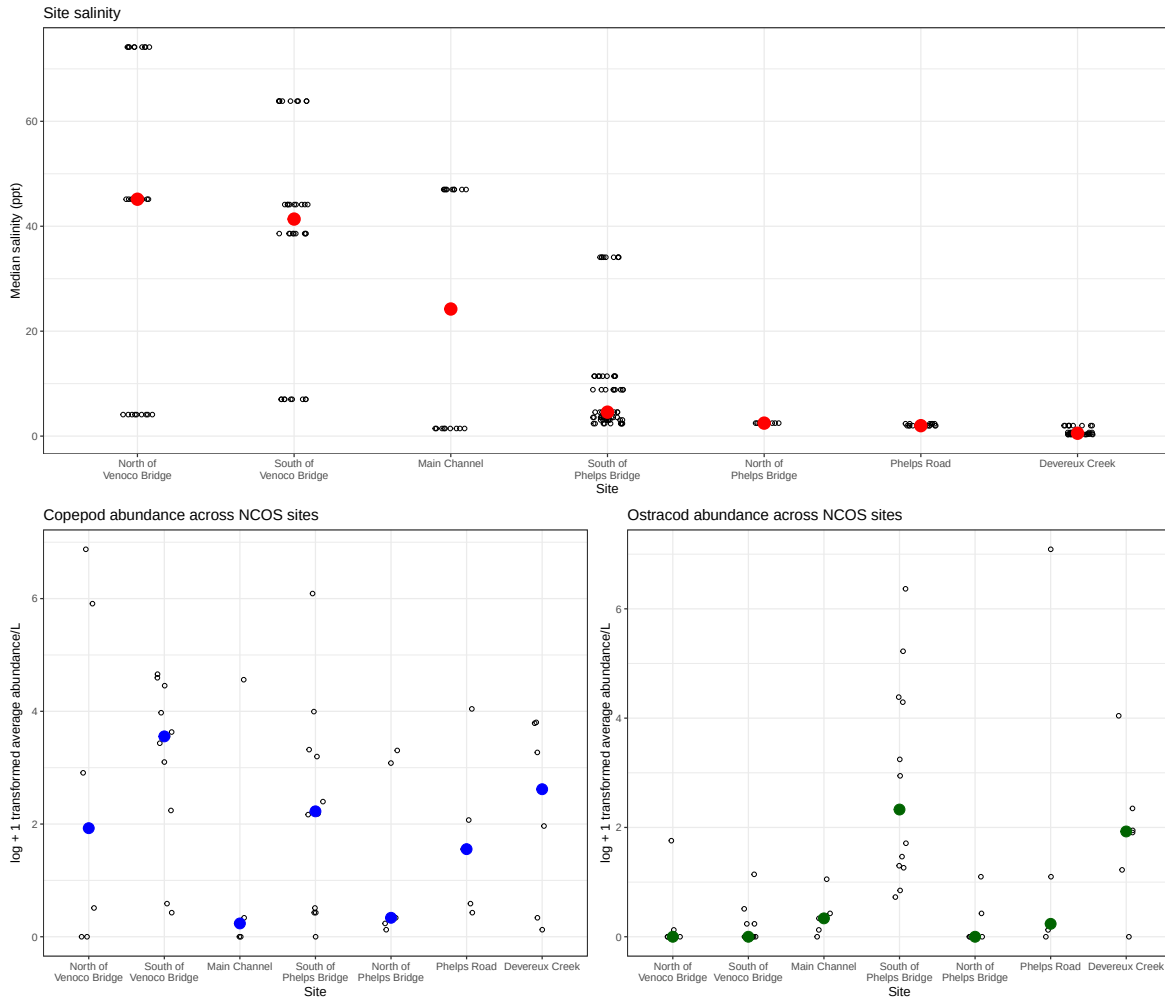
#Adding blue points to show median copepod abundance at each site
stat_summary(geom = "point",
```

```

        fun = "median",
        size = 4,
        color = "blue") +
#Wrapping long site labels on the x-axis
scale_x_discrete(labels = label_wrap(15)) +
#Relabeling axes and adding a title
labs(x = "Site",
      y = "log + 1 transformed average abundance/L",
      title = "Copepod abundance across NCOS sites") +
#Using a different theme from default
theme_bw()

#Combining salinity, copepod, and ostracod figures into one multipanel plot
salinity / (copepods_plot | ostracods_plot)

```



e. Interpret your figure

Copepod and Ostracod abundance differ across sites, but they do not show the exact same site-level pattern. Copepods have the highest median abundance at South of Venoco Bridge, with Devereux Creek and South of Phelps Bridge also showing relatively higher median values. Ostracods, on the other hand, appear highest at South of Phelps Bridge and Devereux Creek. These observations are visible from the larger colored median points in the lower two panels, which are highest at different sites for each taxon.

Ostracod abundance does not appear to increase simply with higher salinity. The highest Ostracod median values occur at South of Phelps Bridge and Devereux Creek, which are both lower-salinity sites compared to North and South of Venoco Bridge in the salinity panel. This suggests that Ostracod abundance may be more significantly shaped by other

site-level conditions, or that its relationship with salinity is not as direct as a simple positive relationship.

2. Visualizing total abundance as a function of

salinity

a. Plan your figure

- X-Axis: med_sal
- Y-Axis: log + 1 transformed average abundance per liter
- Color: taxon
- Geometry: geom_point()
- Facets: facet_wrap(~ taxon, scales = "free")

b. Wrangle your data

```
#Creating a data frame for visualizing abundance as a function of salinity
taxon_salinity <- aq_ins_clean |>
  #Filtering out rows without salinity values
  filter(!is.na(med_sal)) |>
  #Creating log-transformed abundance and cleaner taxon labels
  mutate(log_ave_lit = log(ave_lit + 1),
         taxon_title = str_to_title(str_replace_all(taxon, "_", " "))) |>
  #Ordering taxa by their maximum log-transformed abundance
  mutate(taxon_title = fct_reorder(taxon_title,
                                   log_ave_lit,
                                   .fun = max,
                                   .desc = TRUE)) |>
  #Keeping only the columns needed for the figure and interpretation
  select(date, site, site_full, taxon_title, ave_lit,
         log_ave_lit, med_sal)

#Displaying 10 random rows from the salinity-abundance data frame
taxon_salinity |>
  slice_sample(n = 10)
```

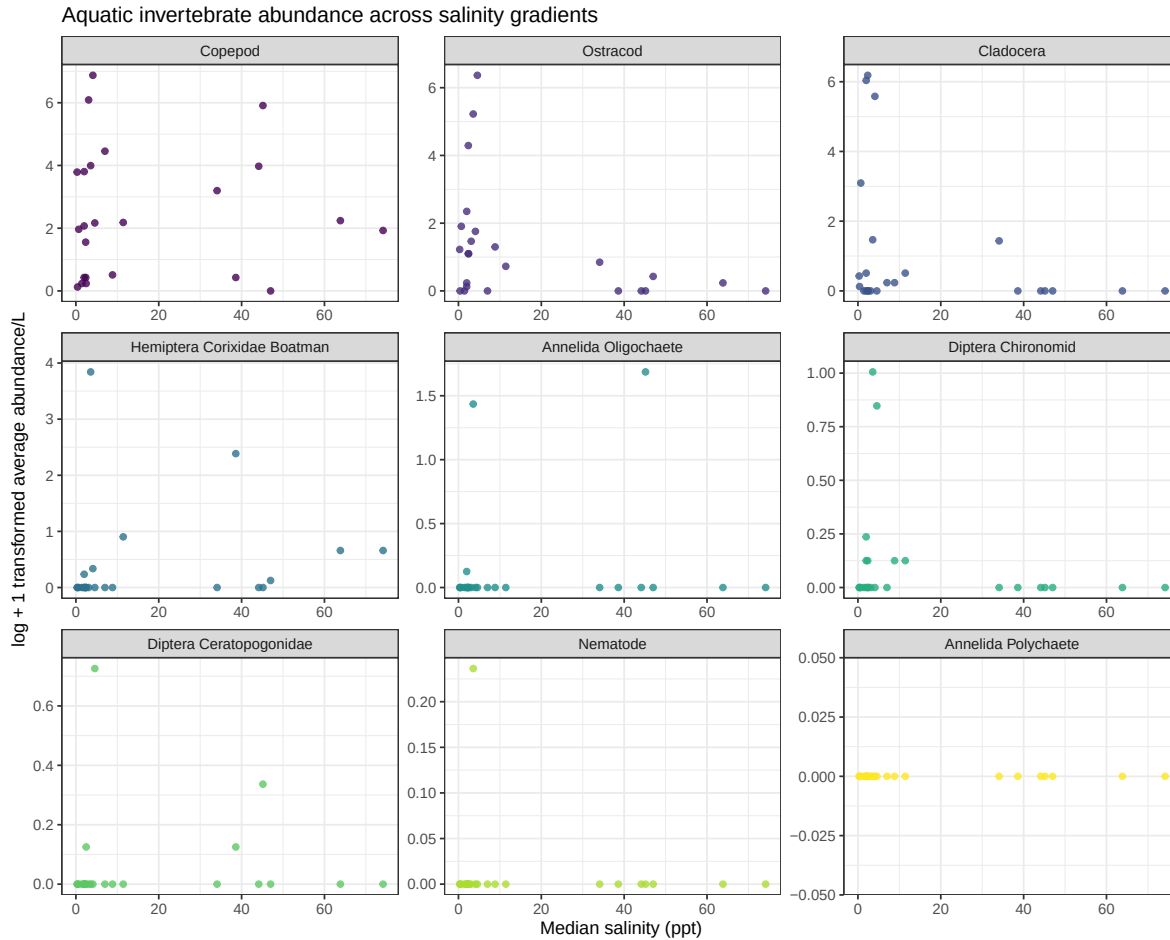
```
# A tibble: 10 x 7
```

date	site	site_full	taxon_title	ave_lit	log_ave_lit	med_sal
------	------	-----------	-------------	---------	-------------	---------

	<dtm>	<chr>	<fct>	<fct>	<dbl>	<dbl>	<dbl>
1	2022-05-06 00:00:00	NMC	Main Chann~	Annelida P~	0	0	1.45
2	2022-05-11 00:00:00	NDC	Devereux C~	Annelida P~	0	0	0.398
3	2022-08-08 00:00:00	NPB2	Phelps Road	Annelida P~	0	0	2.36
4	2022-05-13 00:00:00	NVBR	North of V~	Hemiptera ~	0.933	0.659	74.2
5	2022-01-28 00:00:00	NDC	Devereux C~	Nematode	0	0	2.00
6	2022-01-28 00:00:00	NDC	Devereux C~	Copepod	43.9	3.80	2.00
7	2021-11-23 00:00:00	NPB2	Phelps Road	Ostracod	0.267	0.236	1.99
8	2020-02-21 00:00:00	NVBR	North of V~	Copepod	967.	6.87	4.1
9	2022-02-23 00:00:00	NVBR	North of V~	Ostracod	0	0	45.2
10	2022-05-06 00:00:00	NMC	Main Chann~	Diptera Ce~	0	0	1.45

c. Code your figure

```
#Creating a figure showing abundance as a function of salinity by taxon
ggplot(data = taxon_salinity,
  #Setting x-axis to salinity, y-axis to abundance, and color to taxon
  mapping = aes(x = med_sal,
    y = log_ave_lit,
    color = taxon_title)) +
  #Adding points to show each taxon observation
  geom_point(alpha = 0.8) +
  #Separating each taxon into its own panel with free x- and y-axis scales
  facet_wrap(~ taxon_title,
    scales = "free") +
  #Using non-default colors from the viridis palette
  scale_color_viridis_d() +
  #Relabeling axes and adding a clear title
  labs(x = "Median salinity (ppt)",
    y = "log + 1 transformed average abundance/L",
    title = "Aquatic invertebrate abundance across salinity gradients") +
  #Using a different theme from default
  theme_bw() +
  #Removing the legend because each taxon is already labeled by its panel
  theme(legend.position = "none")
```



d. Interpret your figure

The relationship between abundance and salinity differs across taxa. Copepods show relatively high log-transformed abundance across a wide range of salinity values, including low-salinity sites near 0–10 ppt and higher-salinity sites above 40 ppt, suggesting they occur broadly across the salinity gradient. Ostracods and Cladocerans show stronger abundance at lower salinity values, with the highest points clustered closer to the left side of their panels and lower abundance values appearing at higher salinities. Other taxa, notably Annelida Polychaete as well as Nematode, Diptera Ceratopogonidae, and Diptera Chironomid, mostly have points close to zero across the salinity gradient, so there is not a strong visible relationship between salinity and abundance for those groups. Overall, the faceted plot suggests that salinity may matter more for some taxa than others, but several groups are either rare in the dataset or show mostly low abundance regardless of salinity.